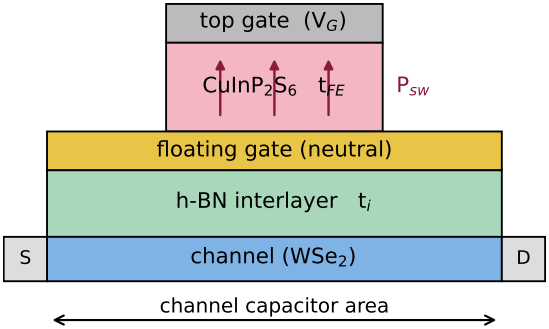
$D_{FE} = D / A_R = \varepsilon_0 \varepsilon_b E_{FE} + P_{sw}$



ferroelectric capacitor area = A_R × channel capacitor area

Stack equations (gate drive x > 0 accumulates holes)

$D = q\,(p - n) + Q_{it}$

$D / A_R = \varepsilon_0 \varepsilon_b\, E_{FE} + P_{sw}$

$x = t_{FE}\, E_{FE} + D\,(1/C_i + 1/C_{FG}) + \psi_s$

Read at constant drain current fixes p, ψ_s:

$MW = t_{FE}\,(\Delta Q_{it}/A_R - \Delta P_{sw})/(\varepsilon_0 \varepsilon_b)$
$\quad + \Delta Q_{it}\,(1/C_i + 1/C_{FG})$

A_R enters ΔP_{sw} through the program step:

$D_p = A_R\, D_{FE,p}$ must be supplied by x_p